# Observer-Patch Holography and the Dark Sector

## Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law

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**Abstract**

In observer-patch holography the Einstein equation sources curvature from the modular-charge stress of the screen under a universal coupling. Every screen source that carries modular charge therefore gravitates, whether or not it carries the electromagnetic readout, and the luminous-only Einstein relation holds exactly when the non-luminous charge vanishes. This is a conditional classification identity for any hypothesized excess, not a derivation that a nonzero dark charge exists. A candidate carrier is the anomalous modular energy on overlap collars, the same finite-stage remainder that the Einstein derivation controls: its rest-frame stress is bounded by the collar recovery defect, vanishes at full recovery, and dilutes as the inverse cube of the scale factor when conserved. The collars that carry it are galactic-scale objects, so a compact source sees only the ambient anomalous density; an isotropic ambient density has no trace-free quadrupole. Its traceful tidal scale in the Solar System is four orders of magnitude below the dimensionally comparable Cassini quadrupole uncertainty, although Cassini does not directly bound the isotropic channel. In the deep regime the anomalous density with linear enclosed mass gives $a_A = \sqrt{a_b a_0}$, flat rotation curves, and the baryonic Tully–Fisher relation $v^4 = GM_b a_0$. A corrected SPARC diagnostic compares the observed acceleration with $a_b + \sqrt{a_b a_0}$; a first comparison against a Tully–Fisher proxy built from the catalogued finite-radius flat speed showed non-overlapping $a_0$ intervals. A matched diagnostic changes to the same baryon-subtracted anomalous-observable family, common cuts, and a bulgeless common sample; its paired interval contains zero, so the two channels are consistent with one fitted constant on this seen sample. This is consistent with, but does not causally isolate, a mixed-proxy explanation and by itself selects no model. The compact-source half of the collar-scale premise is a conditional theorem under an exponential recovery envelope, with a rate in the local record density, and the deep profile with its single constant is characterized, conditional on two named premises, as the unique law satisfying deep-regime scale covariance and quadrature composition of independent sources. All structural statements are machine-checked. A separate source-bound replay of the published empirical full radial-acceleration interpolation reproduces the 153-galaxy parent selection and fits an acceleration scale compatible with the established empirical value, while exposing order-ten-percent weighting sensitivity. Because both the interpolation and dataset were known, that replay strengthens provenance but is not OPH-specific evidence. The magnitude of the anomalous charge and the physical attachment of the envelope are not derived here.

## Claim Boundary

The finite OPH results used here are the rest-frame Einstein relation on its named small-ball premises, the decomposition of the cap modular generator into a bulk part, a central part, and a collar-carried anomalous part, and the controlled bound on that anomalous part [1, 3]. Every theorem and proposition below is an algebraic or analytic consequence of those objects together with the premises named in the text, and each is checked in the Lean formalization [4]. Universal coupling and the collar-scale premise are structure fields or named hypotheses in that formalization; neither is established there. The linear enclosed-mass form is derived inside the formalization from two named premises, deep-regime scale covariance and quadrature source composition, which are themselves structure fields; the exponential recovery envelope behind the saturation theorem is the conditional mixing branch of the collar-recovery claim, taken as a hypothesis.

The paper supplies a finite quotient-invariant receipt localizing the anomalous modular energy on collars, under model premises; the continuum receipt for the physical cap generator, the value of the collar constant, the profile of the anomalous density outside the deep regime, a proof that the collars carrying the anomaly are galactic-scale, and a cosmological likelihood are not supplied. The acceleration constant $a_0$ is a fitted comparison value. No cosmological result carries an OPH falsification verdict.

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## 1 What Sources Gravity in OPH

The conditional Einstein arrow of the spacetime paper ends in

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \tag{1}$$

where $\langle T_{ab} \rangle$ is a supplied local, covariant, conserved stress on the physical branch [1]. Finite positive-null tomography exactly reconstructs a quadratic form from directional charges, but identifying that form with the same local stress additionally requires charge–stress matching, locality and covariance, and a weak Ward law. In the rest frame the relation then follows from three further premises on a small geodesic ball of radius $\ell$: generalized entropy is declared stationary, the bulk entropy variation equals $(8\pi^2\ell^4/15)\,\delta\langle T_{00}\rangle$, and the fixed-volume area variation equals $-(4\pi\ell^4/15)\,\delta G_{00}$. Exact arithmetic gives $\delta G_{00} = 8\pi G\,\delta\langle T_{00}\rangle$.

On that conditional branch, the stress entering this relation is identified with the same modular charge. The screen does not consult the electromagnetic readout of a source before responding to it. That is the universal-coupling premise, and it is the only input the dark-sector question takes from the gravity derivation.

## 2 The Generic Dark Sector

Let a finite family of screen sources be indexed by $i$, each with rest-frame modular-charge contribution $q_i$ and a flag recording whether it carries the electromagnetic readout. Write $Q_{\mathrm{lum}}$ for the sum over flagged sources and $Q_{\mathrm{dark}}$ for the sum over the others, so that $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$.

**Premise 2.1** (Universal coupling)**.** *The stress contraction entering the rest-frame Einstein relation equals $Q_{\mathrm{tot}}$, with no readout-dependent weight.*

**Theorem 2.2** (Geometry responds to all modular charge)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,(Q_{\mathrm{lum}} + Q_{\mathrm{dark}})$.*

**Theorem 2.3** (The luminous-only relation)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,Q_{\mathrm{lum}}$ holds if and only if $Q_{\mathrm{dark}} = 0$.*

**Corollary 2.4** (A geometric excess forces a dark source)**.** *If $\delta G_{00} \neq 8\pi G\,Q_{\mathrm{lum}}$, some source without the electromagnetic readout carries nonzero modular charge.*

**Proposition 2.5** (Attractive sign)**.** *If every non-luminous charge is nonnegative, $Q_{\mathrm{dark}} \geq 0$, and the dark contribution enters the weak-field equation with the sign of baryonic mass.*

*Proofs.* $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$ is a finite-sum identity. Theorem 2.2 substitutes it into the rest-frame relation. Theorem 2.3 cancels $8\pi G > 0$. The corollary is the contrapositive of the sum of zeros, and the proposition is a sum of nonnegative terms. □

These four lines provide a conditional bookkeeping/classification theorem: if universal coupling holds and a geometric excess over the luminous term exists, that excess is called non-luminous modular charge. They do not derive a nonzero $Q_{\mathrm{dark}}$, its abundance, or a physical carrier. Those existence and attachment questions remain separate.

## 3 The Modular-Anomaly Source

On the support-visible branch the cap modular generator has the local form

$$K_C = 2\pi B_C + K_C^{\mathrm{anom}} + \mathrm{const}, \tag{2}$$

with $B_C$ the bulk boost generator and $K_C^{\mathrm{anom}}$ the nonadditive part carried on the overlap collar [1]. At finite regulator stage the Einstein derivation carries this part as a remainder and bounds it:

$$\left|\delta\langle E_C^{(\delta)}\rangle\right| \le C_C\,\eta_\delta, \tag{3}$$

where $\eta_\delta$ is the collar recovery defect and $C_C$ depends only on the collar model and the bounded observable class.

**Definition 3.1** (Anomalous stress and density)**.**

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle, \qquad \rho_A := \frac{1}{c^2}\,\delta\langle T_{00}^{\mathrm{anom}}\rangle. \tag{4}$$

The dark-sector identification is that this anomalous modular energy is the non-luminous charge of Section 2. It carries no electromagnetic readout by construction: it is the part of the modular generator that no local record exposes. Three consequences follow from (3) and (4).

**Theorem 3.2** (Bounded source)**.** $\left|\delta\langle T_{00}^{\mathrm{anom}}\rangle\right| \le 15\,C_C\,\eta_\delta/(8\pi^2\ell^4)$.

**Theorem 3.3** (Full recovery switches the source off)**.** *If $\eta_\delta = 0$ then $\delta\langle T_{00}^{\mathrm{anom}}\rangle = 0$. Along any family with fixed $\ell$, bounded $C_C$, and $\eta_\delta \to 0$, the anomalous stress tends to zero.*

**Theorem 3.4** (Cold scaling)**.** *A conserved anomalous charge in a comoving cell of physical volume $\propto a^3$ has density $\propto a^{-3}$.*

*Proofs.* Theorem 3.2 multiplies (3) by the positive kernel inverse. Theorem 3.3 is the case $\eta_\delta = 0$ of the bound and a squeeze along the family. Theorem 3.4 is $\rho\, a^3 = Q$. $\square$

The split (2) has a machine-checked finite counterpart. On a finite carrier of microstates labelled bulk or collar, with strictly positive weights, a declared boost datum vanishing on the collar, and modular generator $K = -\log w$, the decomposition $K = 2\pi B + K^{\mathrm{anom}} + c\,\mathbf{1}$ exists with $c$ the bulk-weighted mean of $K - 2\pi B$, is unique among splits whose anomalous part has vanishing bulk mean, and is invariant under the additive-constant quotient $w \mapsto e^{-t}w$. When the bulk weights are exactly Gibbs for $2\pi B$, the anomalous part vanishes on the bulk and is carried entirely on the collar, and its expectation is the remainder bounded in (3). The finite carrier, the boost datum, the normalization clause, and the Gibbs condition are model premises; the receipt is the finite counterpart of (2), at exactly that strength.

In the weak-field limit the anomalous density enters the Poisson equation beside the baryons,

$$\nabla^2\Phi = 4\pi G\,(\rho_b + \rho_A), \tag{5}$$

with the attractive sign of Proposition 2.5 whenever the anomalous charge is nonnegative.

## 4 Galactic-Scale Collars and the Solar System

The collars in (2) are overlap collars between observer patches, and the recovery defect $\eta_\delta$ measures how far the repaired record on a collar falls short of determining its complement. Where records are dense and repair saturates, the defect is small; where the screen is sparsely recorded, over the settled outskirts of a galaxy, the defect is carried on long-lived cuts. The long-lived cut has a finite machine-checked counterpart in the consensus-layer provenance package [2]: a freshly injected overlap mismatch that no later commit consumes persists at full strength through every stage of every continuation, so an unconsumed cut is long-lived by theorem. The anomalous charge is therefore a property of the record state of a region, and the region in question is the galaxy.

**Premise 4.1** (Collar scale)**.** *The collars carrying $\delta\langle E_C^{(\delta)}\rangle$ are galactic-scale settled cuts. A compact source inside the galaxy does not generate anomalous charge of its own; it sits in the ambient anomalous density $\rho_A$ of its neighbourhood.*

The compact-source half of this premise is a conditional theorem. The collar-recovery claim of the spacetime paper [1] bounds the defect on its conditional mixing branch by an exponential envelope, which we write in the local record density $\rho$ as

$$\eta_\delta \le \kappa\, B\, \exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{6}$$

with $\kappa$ the mixing prefactor, $B$ the boundary size in UV cells, $r_0$ an offset, $\zeta$ the decay constant, and $\delta\rho^{1/3}$ the collar depth counted in UV record spacings.

**Theorem 4.2** (Compact-source saturation)**.** *Under the envelope (6), along any family of collars with fixed radius $\ell$ and bounded collar constant $C_C$, the recovery defect and the anomalous stress tend to zero as the depth grows at fixed record density, and as the record density grows at fixed positive depth. Quantitatively,*

$$\big|\delta\langle T_{00}^{\mathrm{anom}}\rangle\big| \le \frac{15\, C_C}{8\pi^2\ell^4}\,\kappa B\, \exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{7}$$

*and the envelope bound falls below $\varepsilon$ exactly when $\delta\rho^{1/3} > r_0 + \zeta\log(\kappa B/\varepsilon)$. Where records are dense, a source generates no persistent anomalous charge of its own.*

The delimitation is exact in both directions and machine-checked. The identically zero defect satisfies every envelope hypothesis, so the envelope never forces a nonzero defect: that the galactic-scale settled cuts carry an order-one defect is a premise, not a consequence. And an explicit sparsely recorded instance sits at the bound with an order-one defect and strictly positive stress, so the same hypotheses that screen dense regions admit a nonzero dark source on sparse cuts. What is not established is that physical collars satisfy (6): the envelope is the conditional mixing branch of the collar-recovery claim, taken here as a hypothesis.

The delimitation has a positive counterpart in which the defect is computed rather than admitted. In the erasure-record model, a source bit perfectly correlated with its complement and a collar record carrying the bit with probability $p$, the minimum prediction error over all predictors, deterministic or randomized, is exactly $(1-p)/2$: an unrecorded cut carries an order-one defect necessarily, a saturated record carries none, and the defect falls below $\varepsilon$ exactly when $p > 1 - 2\varepsilon$, the mechanism-side counterpart of the settled-cut threshold. Composed with Theorem 3.3 the sparse instance carries anomalous stress $15/(16\pi^2)$ exactly. The identification of the model defect with $\eta_\delta$ is a modeling premise, and which record density a galactic-scale settled cut realizes is open; the model shows the mechanism class exists, with an exact defect-density law.

Under Premise 4.1 the Solar System is a test of the ambient density alone.

**Proposition 4.3** (An isotropic ambient density has no quadrupole)**.** *The potential of a locally uniform density is $(2\pi G\rho_A/3)\,r^2$. Its projection on the quadrupole weight $P_2(\cos\theta)$ vanishes, $\int_{-1}^{1}(\mu^2 - \frac{1}{3})\,\mathrm{d}\mu = 0$, so the quadrupole coefficient $Q_2$ is zero. The only footprint is the isotropic tidal scale $4\pi G\rho_A/3$.*

With the local dark density $\rho_A \simeq 6.8\times10^{-22}\,\mathrm{kg\,m^{-3}}$, the value inferred from Galactic kinematics [11], the tidal scale is

$$\frac{4\pi G\rho_A}{3} = 1.9\times10^{-31}\,\mathrm{s}^{-2}, \tag{8}$$

For a dimensional scale comparison, the Cassini trace-free quadrupole result is $Q_2 = (1.6\pm1.8)\times 10^{-27}\,\mathrm{s}^{-2}$ [10], four orders of magnitude above the isotropic scale. It is not a direct bound on the traceful isotropic channel, whose quadrupole is identically zero. The anomalous mass enclosed within ten astronomical units is $5\times10^{-15}\,M_\odot$. Both numbers are machine-checked inequalities in the formalization. The Solar System constrains the premise of Section 4 and nothing else in the construction.

## 5 The Deep-Regime Profile

Around a baryonic mass $M_b$, the codimension-one collar count of the microphysics paper [3] gives the number of settled cuts enclosing the source within radius $r$ as proportional to $r$. Rather than positing the enclosed-mass profile as a functional form, the deep regime is characterized by two premises, each independently falsifiable.

**Premise 5.1** (Deep-regime scale covariance)**.** *In the deep regime the enclosed anomalous mass is degree-one homogeneous in the radius: $M_A(M_b,\lambda r) = \lambda\,M_A(M_b,r)$ for every $\lambda > 0$. A sparsely recorded deep cut retains no length scale, so rescaling the radius rescales the enclosed mass by the same factor.*

**Premise 5.2** (Quadrature composition)**.** *The anomalous amplitudes of independent baryonic sources compose in quadrature: $M_A(m_1+m_2,r)^2 = M_A(m_1,r)^2 + M_A(m_2,r)^2$. The collar defects of distinct sources are independent, their variances add, and the gravitating anomalous charge is the amplitude of the unrecovered record fluctuation.*

**Theorem 5.3** (Profile characterization)**.** *Let $M_A(M_b,r)$ be nonnegative, monotone in the source, somewhere nonzero, and satisfy Premises 5.1 and 5.2. Then there is a unique constant $a_0 > 0$ with*

$$M_A(r) = r\sqrt{\frac{M_b a_0}{G}}, \qquad \rho_A(r) = \frac{1}{4\pi r^2}\sqrt{\frac{M_b a_0}{G}}. \tag{9}$$

*Proof.* Scale covariance factorizes $M_A(M_b,r) = r\,f(M_b)$. Quadrature composition makes $f^2$ additive on positive sources, monotonicity makes it monotone, and a function additive and monotone on the positive reals is linear: $f^2 = \lambda M_b$ with $\lambda > 0$ by nondegeneracy. Setting $a_0 = \lambda G$ gives (9); uniqueness follows by evaluating at $M_b = r = 1$. Each premise is load-bearing: an additive law

$M_A \propto rM_b$ satisfies everything except quadrature composition, and an $r^2$ law everything except scale covariance. □

**Corollary 5.4** (Linear enclosed mass)**.** *The deep-regime density is the shell derivative of the enclosed mass, $M_A'(r) = 4\pi r^2 \rho_A(r)$, with $M_A$ linear in $r$ and the coefficient $\sqrt{M_b a_0/G}$ forced by Theorem 5.3.*

Deep-limit scale invariance is the known symmetry of the modified-dynamics limit [8], and square-root mass laws appear in entropic-gravity proposals [9]. The characterization here derives the full profile with its unique constant from the two premises jointly, and it reads the square root as the variance additivity of independent collar defects under source composition.

Premise 5.2 itself follows from a more primitive per-cut model. Suppose the $nr$ nested cuts within radius $r$ each carry anomalous amplitude $\sqrt{V(M_b)}$, the standard deviation of the collar defect sourced by the baryons, with the variances of independent sources adding and the variance monotone in the source. Then the variance function is forced linear, $V(M_b) = cM_b$ with a unique $c > 0$, quadrature composition is a theorem rather than a premise, the induced law satisfies Theorem 5.3, and its constant is exactly

$$a_0 = G\, n^2 c. \tag{10}$$

The dictionary is exact but determines only the product $n^2c$: two models with different cut densities and equal $n^2c$ induce the same enclosed-mass law, machine-checked. The magnitude question is therefore the value of the collar density and of the per-cut variance constant; neither is derived here.

One sufficient realization of both premises rests one level deeper. Variance additivity follows from probabilistic independence: for atomic defect contributions with finite second moments, pairwise independence, and a common per-atom variance on one probability space, the variance of the compound defect of every finite atom family is the sum of the per-atom variances, so disjoint-union composition is additive and $V(m) = cm$ on the modeled masses. Independence is stronger than necessary; pairwise zero covariance and other dependent but uncorrelated constructions remain viable. An exact linear-asymptotic cut count follows from one rigid geometry: a cut family with a positive first cut, consecutive gaps of at least $s$, and every length-$s$ interval meeting a cut is exactly the arithmetic progression with spacing $s$, its count obeys $r/s-1 \le N(r) \le r/s+1$, and the counted enclosed mass deviates from the linear law by at most one per-cut amplitude at every radius, with $n = 1/s$. A perfectly correlated pair and a one-cut family show only that the conclusions do not follow after deleting every decorrelation or density condition. They do not make independence or exact spacing necessary. Irregular or ergodic cut families with asymptotic density and sublinear count error remain possible. The exact $nr$ per-cut law is the continuum or mean extension; the discrete count above differs from it by a bounded error.

**Theorem 5.5** (Deep radial-acceleration relation)**.** *The anomalous acceleration of the profile* (9) *is*

$$a_A(r) = \frac{GM_A(r)}{r^2} = \frac{\sqrt{GM_b a_0}}{r} = \sqrt{a_b(r)\, a_0}, \qquad a_b = \frac{GM_b}{r^2}. \tag{11}$$

**Theorem 5.6** (Flat rotation curve and baryonic Tully–Fisher relation)**.** *The circular speed supported by $a_A$ satisfies $v^2 = a_A r = \sqrt{GM_b a_0}$, independent of $r$, and hence*

$$v^4 = GM_b\, a_0. \tag{12}$$

*Proofs.* $GM_A/r^2 = G\sqrt{M_b a_0/G}/r = \sqrt{GM_b a_0}/r$, and $\sqrt{GM_b a_0}/r = \sqrt{(GM_b/r^2)a_0}$ for $r > 0$. Multiplying by $r$ removes the radius; squaring gives (12). $\square$

The exponents 1/2 in (11) and 4 in (12) are the parameter-free content of the profile. The constant $a_0$ collects $C_C$, the collar density, and the small-ball normalization; none of these is fixed by the source, so $a_0$ enters every comparison as a fitted value. The deep regime is the range $a_b \ll a_0$, equivalently $r \gg r_M = \sqrt{GM_b/a_0}$. If the deep expression is algebraically extended to the matching radius, both accelerations equal $a_0$ there and the anomalous term dominates outside $r_M$; those algebraic statements are machine-checked. The premises are asserted only in the deep regime, so they do not determine the physical transition or the response inside $r_M$.

## 6 Lensing and the Relativistic Source

The anomalous charge must enter (1) through one stress tensor; Premise 2.1 supplies no readout-dependent coupling. That fact does not imply equality of masses inferred from null geodesics and timelike orbits. Those observables contract different components of the metric response and can differ in the presence of pressure, anisotropic stress, or gravitational slip. No relativistic refinement limit is supplied that turns the weak-field density (9) into a full $T_{ab}^{\text{anom}}$. A lensing law and the relation between lensing and dynamical mass therefore remain open physical constructions.

## 7 Comparison with Rotation-Curve Data

Theorems 5.5 and 5.6 are compared with the SPARC sample of 175 disk galaxies with $3.6\,\mu$m photometry [6] under the declared inclusive-error deep-law diagnostic: quality flag at most 2, inclination at least 30°, velocity error at most ten percent, and stellar mass-to-light ratios 0.5 for disks and 0.7 for bulges. This leaves 2700 points in 149 galaxies. The strict published-cut replay described below is kept separate. Both computations are released with the paper [7, 5].

**Source-bound full-RAR calibration.** Separate from the conditional OPH deep law, a deterministic replay fits the published empirical interpolation

$$g_{\text{obs}} = \frac{g_{\text{bar}}}{1 - \exp[-\sqrt{g_{\text{bar}}/a_0}]}$$

to a committed CDS/VizieR snapshot whose provenance, catalogue description, and two table files are pinned by SHA-256. The catalogue cuts reproduce the published 153-galaxy parent selection. The strict point cuts retain 2696 points in 147 contributing galaxies, three more than the published 2693-point census. These counts are distinct from the preceding 2700-point, 149-galaxy deep-law diagnostic: that diagnostic uses the inclusive cut $dV/V \leq 0.1$, whereas this full-RAR replay uses the strict cut $dV/V < 0.1$ and starts from the 153-galaxy parent set. With disk and bulge mass-to-light values 0.5 and 0.7, the unweighted point-level log-residual fit gives

$$a_0 = (1.1613 \pm 0.0802) \times 10^{-10}\,\mathrm{m\,s^{-2}}, \qquad \sigma_{\log_{10} g} = 0.1327 \text{ dex},$$

where the width is the standard deviation of a galaxy bootstrap. Equal total weight per contributing galaxy shifts $a_0$ by $-9.58\%$; weighting only by inverse fractional velocity variance shifts it by

+15.49%. These are estimator-sensitivity controls, not calibrated likelihood alternatives. The interpolation and acceleration scale were known before this replay, no OPH source value enters the fit, and distance, inclination, gas, stellar population, intrinsic scatter, and correlation uncertainties are not jointly marginalized. The result certifies public-data custody and arithmetic, not independent evidence for OPH [7, 5].

**Radial acceleration.** On fixed absolute subsets $g_{\rm bar} < f a_{\rm ref}$, with the comparison-only value $a_{\rm ref} = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$ fixed before fitting, the density profile predicts the total acceleration $g_{\rm obs} = g_{\rm bar} + \sqrt{g_{\rm bar} a_0}$. The fixed-exponent diagnostic gives

$$a_0 = 0.843 \times 10^{-10}, \quad 0.875 \times 10^{-10}, \quad 0.924 \times 10^{-10}\ {\rm m\,s^{-2}}$$

for $f = 0.3$, $0.1$, and $0.03$, with 1519, 960, and 203 points in 141, 115, and 40 galaxies and rms scatter between 0.14 and 0.17 dex. Fitting an anomalous power inside the total model, using every retained point rather than selecting on the sign of the noisy excess, gives exponents 0.454, 0.461, and 0.239. Galaxy-cluster bootstrap intervals contain 1/2 for the first two cuts; the deepest cut is sparse and its interval is wide. The computation uses equal retained-point weighting and does not marginalize distance, inclination, stellar-population, or correlated rotation-curve uncertainties, so it is a diagnostic rather than a full measurement likelihood.

**Baryonic Tully–Fisher.** For the 123 galaxies with a positive measured flat velocity that pass the catalogue quality and inclination cuts (without applying the rotation-point fractional-velocity-error cut), with $M_b = 0.5L_{[3.6]} + 1.33M_{\rm HI}$, the fixed-exponent normalization $a_0 = v_f^4/(GM_b)$ gives $a_0 = 1.55 \times 10^{-10}\,{\rm m\,s^{-2}}$ with 0.26 dex scatter, and the free exponent from an ordinary least-squares fit of $\log v_f$ on $\log M_b$ is $3.75 \pm 0.10$.

**One constant.** Theorems 5.5 and 5.6 share one $a_0$. A first comparison estimated the two constants from mismatched observables: the radial-acceleration diagnostic above against the fixed-exponent normalization $a_0 = v_f^4/(GM_b)$ built from the catalogued $V_{\rm flat}$. That mixed diagnostic gave paired parent-galaxy bootstrap intervals $[0.730, 1.037] \times 10^{-10}\,{\rm m\,s^{-2}}$ and $[1.405, 1.723] \times 10^{-10}\,{\rm m\,s^{-2}}$ with a log-ratio 95 percent interval of $[0.180, 0.317]$ dex. The catalogued $V_{\rm flat}$ is read at a finite measurement radius where the baryonic contribution to $V^2$ has not been subtracted, and that bias has the direction of the mismatch.

The matched diagnostic removes the mixed observable by comparing the same anomalous observable in both channels: the baryon-subtracted squared speed $v_A^2 = v_{\rm obs}^2 - v_{\rm bar}^2$ at the outermost measured radius, restricted to galaxies whose outermost point passes the same fixed absolute deep cut $g_{\rm bar} < 0.1\,a_{\rm ref}$ used at $f = 0.1$. Galaxies with any nonzero retained bulge component are excluded because table 1 supplies only total $L_{[3.6]}$, not the disk/bulge luminosity split needed for a consistent mass denominator. On the resulting 97-galaxy common set the second channel gives $a_0 = 0.769 \times 10^{-10}\,{\rm m\,s^{-2}}$, while the radial-acceleration fit on the same common set gives $0.853 \times 10^{-10}\,{\rm m\,s^{-2}}$; the paired $\log_{10}$ ratio is $-0.045$ with galaxy-level bootstrap 95 percent interval $[-0.117, +0.026]$, which contains zero. The full $f = 0.1$ channel-A fit, before the common-set restriction, is $0.875 \times 10^{-10}\,{\rm m\,s^{-2}}$. A synthetic sample with a known $a_0$ recovers a unit paired ratio through the same pipeline, and an independent script re-derives every quoted number from the released receipt. The combined matched-analysis changes remove the displacement and are consistent with a mixed-proxy explanation, but they also change the radius, subtraction, sample, and combination rule together; causal attribution requires staged one-change-at-a-time ablations. The deep-regime profile is consistent with one fitted constant across the matched channels on this seen sample. Consistency on seen data selects nothing: this is a postdiction diagnostic with declared mass-to-light ratios and cuts, and a physical verdict requires the joint nuisance and measurement-error likelihood named in Section 9. The generic modular-charge dark-sector statements do not depend on this profile fit.

**Penalized profile objective.** A declared-convention objective is released with the paper: a per-point Gaussian rotation-curve data term for $g_{\rm obs} = g_{\rm bar} + \sqrt{g_{\rm bar} a_0}$ with per-galaxy profiled nuisances (disk mass-to-light under a log-normal prior centered at 0.5, distance and inclination at their catalogued errors), the variance-normalization term kept in the objective, and a declared equal-correlation intra-galaxy covariance family scanned over $\rho \in \{0, 0.2, 0.4, 0.6\}$. Because the mass-to-light constraint is an astrophysical prior rather than an identified auxiliary-data likelihood, this is a penalized objective, not a maximum-likelihood analysis. Its delta-objective threshold sets are uncalibrated sensitivity contours rather than confidence intervals, and every disconnected component is reported. On the $f = 0.1$ deep subset the objective minimum moves from 0.794 to $0.582 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ as $\rho$ runs from 0 to 0.6, a 0.135-dex shift and a large unresolved sensitivity within the scanned covariance family. Other unquantified systematics prevent a total systematic ranking, and no single $\rho$ is selected. The data quadratic form divided by the nominal fixed-nuisance denominator lies between 3.6 and 16.4 on every subset and correlation value, so the declared error family, which carries no intrinsic-scatter term, understates the observed scatter. The paired comparison against the matched baryon-subtracted channel contains zero at every cut and correlation value, but it is only partially paired: channel A is re-minimized in each bootstrap replicate while channel-B nuisance values are held at their full-sample point estimates. The source does not fix $a_0$; the released receipt records measurement-side sensitivity contours only. They are not acceptance or rejection bands for a future source derivation, which must be scored against a prospectively frozen, calibrated covariance and intrinsic-scatter likelihood. An independent script re-derives the receipt byte for byte, a synthetic sample with known $a_0$ recovers it through the same pipeline as a regression sanity check, and mutation guards cover the subtraction sign, the cut, the mass-to-light encoding, and the covariance algebra.

# 8 Cosmological Scaling

If the total anomalous charge in a comoving region is conserved, which holds whenever the collar family of the region is stable under the expansion, Theorem 3.4 gives $\rho_A \propto a^{-3}$: the anomalous medium has the homogeneous dilution law of a pressureless component. This does not derive its pressure perturbations, anisotropic stress, sound speed, or clustering. Premise 4.1 associates its carrier with galactic collars but supplies no perturbation dynamics. The abundance $\Omega_A$, initial data, transfer functions, and the claim that it clusters with galaxies therefore remain open.

# 9 Required Physical Constructions

1. **Collar scale.** The compact-source half of Premise 4.1 is Theorem 4.2, conditional on the exponential envelope (6). What remains is the physical attachment: a proof that physical collars satisfy the mixing branch behind the envelope, and a positive mechanism for the order-one defect on galactic-scale settled cuts, which the envelope provably does not force.

2. **Magnitude.** The normalization of $\delta\langle E_C^{(\delta)}\rangle$ to repair conservation on the Ward/Bianchi branch is not derived. The per-cut dictionary (10) reduces the question to the collar density $n$ and the per-cut variance constant $c$: a repair-conservation normalization would fix $c$, and the codimension-one collar count of the microphysics paper would fix $n$.

3. **Profile.** Theorem 5.3 reduces the profile to Premises 5.1 and 5.2; what remains is deriving those two premises from the collar geometry of a point source, and extending the profile through the transition to the Newtonian regime.

4. **Localization.** The finite quotient-invariant receipt of Section 3 identifies the collar operator with the bounded remainder on a finite carrier under the Gibbs-on-bulk condition; what remains is the continuum identification for the physical cap generator, whose modular flow the finite boost datum only declares.

5. **Abundance.** The comoving anomalous charge of the early screen, the resulting $\Omega_A$, and the joint galaxy, cluster, lensing, CMB, and growth likelihood are not derived. Within rotation curves, the released penalized profile objective is a conditional sensitivity diagnostic: a calibrated intra-galaxy covariance, an intrinsic-scatter model, and a source value of $a_0$ are the open measurement-side constructions.

# 10 Conclusion

Under universal coupling the screen gravitates according to its total modular charge, yielding a conditional classification theorem: any geometric excess over the luminous term is represented as non-luminous modular charge. This does not establish that the excess is nonzero or attach a physical carrier. The anomalous modular energy on overlap collars is a candidate carrier supplied by the finite Einstein derivation. It is bounded by the recovery defect, absent at full recovery, and cold when conserved. On galactic-scale collars its Solar-System footprint has zero trace-free quadrupole and a small traceful isotropic tidal scale; the four-order Cassini comparison is dimensional, because that measurement does not directly bound the isotropic channel. Its deep-regime profile gives the radial-acceleration relation, flat rotation curves, and the baryonic Tully–Fisher relation with one constant. Under the declared fixed-mass-to-light diagnostic, the SPARC radial-acceleration and matched anomalous-speed channels agree on that constant within the paired bootstrap interval, on seen data, and the released penalized profile objective retains consistency at every scanned intra-galaxy correlation value; its contours are uncalibrated and some are disconnected. The separate source-bound full-RAR replay reproduces the published parent-galaxy selection and acceleration scale, while its simple weighting controls shift the fitted scale by roughly ten to fifteen percent; it is a custody and estimator-sensitivity result, not OPH evidence. A calibrated covariance and intrinsic-scatter treatment stays required for a physical verdict; the correlation scan is a large unresolved sensitivity, not a ranking of the total systematic budget. The compact-source half of the collar-scale premise is a conditional saturation theorem with an exponential rate in the local record density, and the profile with its unique constant is characterized by deep-regime scale covariance and quadrature composition of independent sources. The physical attachment of the envelope, the mechanism selecting galactic-scale cuts, and the magnitude of the charge are the work that turns the remaining premises of Sections 4 and 5 into derivations.

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